

Pre-processing & Feature Engineering Report

AI Forensic Triage Tool: Predicting Shooting Incidents in Boston

Capstone Project — DSE 6311

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Recap Background & Question

Forensic crime laboratories across the United States, including the Massachusetts State Police Crime Laboratory, operate under intense pressure. The volume of cases continues to grow while staffing and equipment resources remain limited. Ballistics analysis, DNA testing, firearm examinations, and other specialized forensic work are extremely time-consuming processes that can take weeks or even months to complete. Every hour spent on a low-priority case delays justice for victims of serious violent crimes.

In Boston, the situation is particularly challenging. The Boston Police Department responds to tens of thousands of reported incidents each year, yet only a small fraction of those incidents actually involve a shooting. Despite this, every reported crime with a potential firearm component must be treated as a possible shooting until proven otherwise. This creates a significant bottleneck: the crime lab must decide which cases deserve immediate forensic attention and which can safely wait. Without an objective, data-driven way to prioritize, valuable laboratory resources are often spread too thin, slowing investigations and reducing the overall effectiveness of the criminal justice system.

This capstone project directly addresses that real-world challenge by developing an AI Forensic Triage Tool. The goal is to help forensic analysts and police leadership quickly identify which reported incidents are most likely to involve a shooting so that limited laboratory resources such as ballistics testing, shell casing analysis, and DNA processing can be focused on the cases with the highest public safety impact.

The central research question guiding the entire project is:

Can incident features (time of day, location, district, offense type proxies) combined with neighborhood demographics accurately predict whether a reported crime will involve a shooting?

We hypothesized that certain patterns would emerge strongly from the data. Specifically, we predicted that nighttime incidents occurring in higher-poverty districts would show a significantly

elevated probability of involving a shooting at least 35% higher compared to daytime incidents or those in lower-poverty areas. This hypothesis is grounded in both criminological research and the lived experience of Boston police officers and forensic examiners who have long observed that violence tends to concentrate in specific times and neighborhoods.

By building a predictive model that can assign a reliable shooting probability score to each new incident, the project aims to give the Massachusetts State Police Crime Laboratory and the Boston Police Department a practical, explainable tool they can use in daily operations. Ultimately, the hope is that faster triage will lead to quicker forensic results, stronger cases, and faster justice for victims and communities across Boston.

Methods

The preprocessing and feature engineering phase was designed with two main goals in mind: (1) prepare a clean, high-quality dataset that a supervised model could learn from effectively, and (2) create features that would be both statistically powerful and easily understandable by the Massachusetts State Police Crime Laboratory and Boston Police Department stakeholders. Every decision was made with interpretability and real-world usability in mind rather than maximizing model complexity.

1. Data Acquisition and Initial Cleaning

The project began with the publicly available Boston Police Crime Incident Reports dataset (2023–present), which contained approximately 239,371 rows at the start of analysis. The raw data was loaded using the pandas library from a locally stored parquet file for efficiency.

Initial cleaning steps included:

- Removing or appropriately imputing missing values in critical columns such as latitude, longitude, district, and offense code group.
- Standardizing date-time fields and extracting the hour of the incident.
- Filtering out clearly erroneous records (e.g., coordinates falling well outside the Boston metropolitan area).
- Creating a clean binary target variable SHOOTING (1 = shooting involved, 0 = no shooting).

All cleaning operations were documented in Notebook 01 (01_data_wrangling.ipynb) with detailed comments so the process remains fully reproducible.

2. Feature Engineering

After basic cleaning, we engineered several new features that capture meaningful behavioral and contextual patterns:

- **Circular time encoding:** Instead of treating the hour of the day as a simple linear number (0–23), we created `hour_sin` and `hour_cos`. This prevents the model from treating 11 PM and 1 AM as far apart when they are actually close in the daily cycle.
- **Binary indicator variables:**
 - `is_night`: 1 if the incident occurred between 8 PM and 6 AM (a period historically associated with higher violence).
 - `is_weekend`: 1 if the incident occurred on Saturday or Sunday.
 - `is_violent`: 1 if the offense code group description contained keywords suggesting violent crime (e.g., Assault, Robbery, Shooting). This proxy was created using string matching on the `OFFENSE_CODE_GROUP` column.
- **Neighborhood socioeconomic context:** We merged district-level poverty rate data from the U.S. Census American Community Survey (ACS) 2020–2024. This added a single numeric column `poverty_rate` that gives the model important contextual information about the economic conditions of the neighborhood where the incident occurred.
- **Categorical encoding:** The `DISTRICT` variable was one-hot encoded into 19 binary columns so the model could learn district-specific risk patterns without assuming any artificial ordering.

These features were chosen deliberately because they are grounded in both criminological literature and practical knowledge shared by police officers and forensic examiners. Each engineered feature is directly interpretable, which is essential for stakeholder trust.

3. Unsupervised Methods Considered

Although the final goal is a supervised prediction model, we performed several unsupervised exploratory steps to better understand the data structure before moving to modeling:

- Correlation analysis and heatmap visualizations were generated in Notebook 02 (02_eda.ipynb) to identify potential multicollinearity and confirm that the engineered features provided independent signals.
- We briefly considered Principal Component Analysis (PCA) as a dimensionality reduction technique for the one-hot encoded district variables. However, after reviewing the results, we decided against using PCA because reducing interpretability would make it much

harder for the crime lab to understand and trust the model's predictions. In a real-world forensic setting, explainability is more valuable than a small gain in computational efficiency.

These unsupervised explorations helped validate our feature choices and guided the final supervised modeling plan.

4. Handling Class Imbalance

The target variable SHOOTING was severely imbalanced (only 0.70% of incidents involved a shooting). To address this, we applied **SMOTE (Synthetic Minority Over-sampling Technique)** exclusively to the training set. This increased the effective shooting rate in the training data to 50% while leaving the test set completely untouched, ensuring a realistic and unbiased evaluation.

5. Plan for Supervised Methods

With the cleaned and engineered dataset now ready (239,371 rows × 24 features), we are well positioned to move into supervised modeling. The plan includes:

- Training an **XGBoost classifier** as the primary model due to its strong performance on tabular data, built-in handling of missing values, and excellent interpretability when combined with SHAP.
- Using **Precision-Recall AUC** as the main evaluation metric because of the severe class imbalance.
- Generating **SHAP values** for every prediction to provide clear, instance-level explanations that the crime lab can use to understand why a particular incident received a high (or low) shooting probability score.
- Performing cross-validation and hyperparameter tuning in the next phase to ensure the model generalizes well.

All preprocessing code, engineered features, and the final prepared dataset are saved in the GitHub repository under data/processed/ and notebooks/. This ensures full reproducibility and allows the stakeholder to trace every step from raw data to final model input.

Results & Brief Interpretations

After completing the full preprocessing and feature engineering pipeline, the dataset was transformed from raw, messy police incident records into a clean, model-ready feature matrix. The final dataset contained 239,371 rows and 24 engineered features. This represents a substantial improvement in data quality and usability compared to the original raw file.

Key Outcomes from Preprocessing

- Missing values in critical columns (latitude/longitude, district, offense code group) were handled appropriately without introducing bias.
- All date-time fields were standardized, allowing reliable extraction of temporal patterns.
- The binary target variable SHOOTING was created cleanly, confirming the severe class imbalance: only 0.70% of incidents involved a shooting.

Feature Engineering Results

The engineered features provided rich, interpretable signals:

- Circular time encoding (hour_sin and hour_cos) successfully captured the cyclical nature of the day, preventing the model from treating midnight and 11 PM as distant.
- Binary flags (is_night, is_weekend, is_violent) highlighted well-known risk periods and offense types.
- The merged district-level poverty_rate from U.S. Census ACS data added important neighborhood context that would otherwise have been missing.
- One-hot encoding of the 19 police districts allowed the model to learn location-specific risk patterns.

Class Imbalance Handling

Because the target was so imbalanced, SMOTE was applied only to the training set. This increased the effective shooting rate in the training data to 50%, while the test set remained untouched at the original 0.70% rate. This approach ensured the model learned from a balanced training environment without contaminating the evaluation.

Exploratory Analysis Highlights

Unsupervised exploration in Notebook 02 revealed several important patterns:

- Nighttime incidents showed a dramatically higher shooting rate (1.59%) compared to daytime (0.31%) — a roughly 5× increase.
- Districts B2, B3, and C11 consistently showed the highest shooting probabilities, aligning with known higher-poverty neighborhoods.
- The correlation heatmap confirmed that the engineered features were not highly collinear, supporting their independent value.

Supervised Modeling Readiness

The final feature matrix was saved as both a parquet file and a CSV for easy use in modeling. An initial XGBoost baseline model trained on this data achieved a Precision-Recall AUC of 0.8327, which is a strong result given the extreme class imbalance.

SHAP analysis on the trained model provided clear interpretability:

- Poverty Rate emerged as the single most influential feature.
- Hour of Day and Is Night were the next strongest predictors, confirming the strong temporal component of risk.
- Specific districts (especially B2 and A7) also contributed meaningfully.

These results strongly support the original hypothesis: nighttime incidents in higher-poverty districts do indeed carry significantly higher shooting risk. The model is now ready for the next phase of hyperparameter tuning, cross-validation, and full stakeholder-facing explainability work.

All results, including the engineered dataset, SHAP plots, and performance metrics, are saved in the GitHub repository under the `models/` and `visualizations/` folders for full transparency and reproducibility.

Discussion & Next Steps

The preprocessing and feature engineering phase has successfully transformed a large, raw, and imperfect police incident dataset into a clean, well-structured, and highly informative feature matrix ready for modeling. This work represents far more than simple data cleaning, it was a deliberate process of translating real-world forensic and policing challenges into features that a model can learn from while remaining understandable to the stakeholders who will ultimately use the tool.

One of the strongest outcomes of this phase is how well the engineered features align with both the research question and the original hypothesis. The creation of `is_night`, circular time encoding, the violent offense proxy, and especially the district-level poverty rate from Census data gave the model meaningful signals that reflect known patterns of violence in Boston. The exploratory analysis in Notebook 02 confirmed that nighttime incidents are roughly five times more likely to involve a shooting than daytime incidents, and that certain districts (particularly B2, B3, and C11) consistently show elevated risk. These findings directly support our hypothesis that nighttime incidents in higher-poverty neighborhoods carry significantly higher shooting probability.

The decision to prioritize interpretability over aggressive dimensionality reduction (such as using PCA on the district variables) was one of the most important choices made during this phase. While PCA could have reduced the number of features slightly, it would have come at the cost of stakeholder trust. Forensic laboratories and police leadership need to understand *why* the model flags a particular incident as high-risk. By keeping features like individual districts, night-time indicator, and poverty rate intact, we ensure that every prediction can be explained in plain language. This focus on explainability will be critical when the tool is presented to the Massachusetts State Police Crime Laboratory.

The use of SMOTE to address the severe class imbalance was also handled thoughtfully. Applying it only to the training set preserved the realistic distribution in the test data, allowing for honest evaluation of model performance. The resulting Precision-Recall AUC of 0.8327 on the baseline model is encouraging and suggests the engineered features are carrying strong predictive signal.

Lessons Learned

This phase highlighted several important lessons. First, domain knowledge remains irreplaceable, features like `is_night` and the poverty rate were far more effective than purely statistical transformations because they reflect real operational realities faced by the crime lab. Second, the time invested in thorough EDA (correlation heatmaps, shooting rate by hour and district, etc.) paid off by guiding smarter feature decisions and preventing us from including redundant or noisy variables. Finally, maintaining a reproducible workflow (clear notebook structure, consistent file

paths, and detailed comments) made the entire process much smoother and will make future updates or collaboration significantly easier.

Limitations

While the current preprocessing is solid, a few limitations remain. The poverty rate is currently calculated at the district level rather than the more granular census tract level due to time constraints and geocoding complexity. In the future, incorporating tract-level socioeconomic data could add even more precision. Additionally, the violent offense proxy relies on keyword matching in the offense code group description; a more sophisticated natural language approach could potentially capture nuances better.

Next Steps – Modeling Plan

With the dataset now fully prepared, the project is ready to move into the core supervised modeling phase. The immediate next steps include:

1. **Model Training and Tuning** — Train the final XGBoost classifier with systematic hyperparameter tuning and cross-validation to ensure robust performance.
2. **Full SHAP Explainability** — Generate comprehensive SHAP dependence plots and force plots to provide detailed, instance-level explanations for stakeholders.
3. **Model Evaluation** — Conduct a thorough evaluation using Precision-Recall AUC, F1-score, and confusion matrices on the untouched test set.
4. **Interactive Dashboard** — Integrate the final model predictions and SHAP values into the existing Tableau dashboard so users can explore risk scores and explanations interactively.
5. **Stakeholder Validation** — Prepare a formal presentation and report for the Massachusetts State Police Crime Laboratory and Boston Police Department to gather feedback on usability and interpretability.
6. **Deployment Considerations** — Explore options for a lightweight deployment (e.g., a simple API or scheduled batch scoring) so the tool can eventually run on new incoming incidents in near real-time.

Overall, the preprocessing and feature engineering work has laid a strong, transparent, and stakeholder-aligned foundation. The model built on this dataset has a high likelihood of delivering practical value to forensic operations in Boston. The next phase will focus on refining the model and delivering an interactive tool that the crime lab can confidently use to prioritize cases and accelerate justice.

Appendix: Data Dictionary (Selected Key Features)

The following table summarizes the most important variables used in the final model. All features were created or transformed during the preprocessing and feature engineering phase to maximize both predictive power and interpretability for forensic stakeholders.

Feature	Type	Description	Rationale / Stakeholder Value
hour_sin / hour_cos	Numeric	Circular encoding of the hour of the incident (0–23)	Captures the cyclical nature of time so midnight is correctly recognized as close to 11 PM
is_night	Binary	1 if incident occurred between 8 PM and 6 AM	Strong predictor of elevated shooting risk; directly actionable for patrol and lab prioritization
is_weekend	Binary	1 if incident occurred on Saturday or Sunday	Weekend patterns differ from weekdays and help refine risk scoring
is_violent	Binary	1 if offense code group suggests a violent crime (e.g., Assault, Robbery)	Proxy for offense severity; helps the lab understand why an incident received a high probability score
poverty_rate	Numeric	District-level poverty rate from U.S. Census ACS 2020–2024	Strongest SHAP feature; provides critical neighborhood context without needing tract-level geocoding
DISTRICT_* (19 columns)	Binary	One-hot encoded police districts (A15, A7, B2, etc.)	Allows the model to learn district-specific risk patterns

Feature	Type	Description	Rationale / Stakeholder Value
			that are well-known to Boston officers
SHOOTING	Binary	Target variable: 1 = shooting involved, 0 = no shooting	Binary classification target; only 0.70% positive cases in raw data
INCIDENT_NUMBER	String	Unique identifier for each incident	Retained for traceability and linking back to original records
Lat / Long	Numeric	Geographic coordinates of the incident	Used for mapping in the Tableau dashboard

Additional engineered variables (not shown in table for brevity) include hour, YEAR, MONTH, and the original raw fields that were retained for reference.

Full Data Dictionary Location

A complete codebook with every column, original source, transformation steps, and summary statistics is available in the GitHub repository:

- reports/Data_Dictionary_Full.md
- data/processed/tableau_ready.csv (final version used for the interactive dashboard)

All preprocessing code, feature engineering steps, and the final prepared dataset can be found in:

- notebooks/01_data_wrangling.ipynb
- notebooks/03_feature_engineering_and_modeling.ipynb

This data dictionary ensures that any stakeholder (or future team member) can fully understand every variable used in the model and the rationale behind each engineering choice.

End of Report

This Pre-processing & Feature Engineering Report documents the complete journey from raw Boston Police data to a clean, model-ready dataset. All code, intermediate files, models, SHAP plots, and the interactive Tableau dashboard are available in the public GitHub repository:

GitHub Repository: <https://github.com/Rick-997/Al-Forensics-Boston-Capstone>

Live Tableau Dashboard:

[AI Forensic Triage Tool – Boston Shooting Risk Predictor](#)

The next phase will focus on final model tuning, comprehensive SHAP explainability, and delivering a production-ready tool for the Massachusetts State Police Crime Laboratory and Boston Police Department.